Business Process Engineering
Week # 8

FALL-2025

Business Waste

- Anything that doesn't increase value in the eye of the customer must be considered 'waste'.
- Every effort should be made to eliminate that waste.
- Toyota Production System (TPS) created a manufacturing framework, which was based on the idea of preserving (or increasing) value with less work.
- There are 8 manufacturing wastes, which have a universal application to businesses today – DOWNTIME

Defects

- Mistakes that require additional time, resources, and money to fix
- Completely eradicating any form of waste is impossible
- Defects can be limited by application of standardized work plans, strict quality controls, full understanding of work requirements and customer needs.
- Some causes of defects include:
  - Poor quality controls
  - Poor repair
  - Poor documentation
  - Lack of standards
  - Weak or missing processes
  - Misunderstanding customer needs
  - Poor design and undocumented design changes

Overproduction

- In some organizations, workers blindly keep producing even when those who receive their output either aren't ready for it or don't need it
- It can tie up significant working capital
- Solution is to establish a reasonable work flow for the benefit of the customer
- Causes of overproduction include:
  - Just-in-case production
  - Unclear customer needs
  - Producing to a forecast
  - Long set-up times
  - Engineering changes
  - Poorly applied automation

Waiting

- Whenever work has to stop for some reason – i) next person in line is overwhelmed; ii) something broke down; iii) waiting for approval or material;
- Solution is to provide adequate staffing to handle the workload at the bottlenecks
- Causes include:
  - Unbalanced workloads
  - Unplanned downtime
  - Long set–up times
  - Producing to a forecast
  - Insufficient staffing
  - Work absences
  - Poor process quality
  - Poor communication

Non-utilized talent

- Not or under-utilizing people's talents, skills and knowledge can have a detrimental effect on an organization
- Companies can experience great benefits when recognizing the value of skills and improvement ideas from all levels of the business and can suffer when not effectively engaging in the process
- Non-utilizing talent can be seen with:
  - Assigning staff to wrong tasks
  - Wasteful admin tasks
  - Poor communication
  - Lack of teamwork
  - Poor management
  - Insufficient training

Transportation

- Wastes caused by moving things around
- Too much transportation tends to increase costs, wastes time, increases the likelihood of product damage and deterioration, and can result in poor communication
- Limiting transportation waste can be done by simplifying processes, repairing physical layouts and handling products less often
- Causes include:
  - Poor plant/office layout
  - Unnecessary or excessive steps in the process
  - Maligned process flow
  - Poorly-designed systems

Inventory excess

- This waste occurs when there is supply in excess of real customer demand, which masks real production
- Causes include:
  - Overproduction and buffers
  - Poor monitoring systems
  - Mismatched production speeds
  - Unreliable suppliers
  - Long set-up times
  - Misunderstood customer needs

Motion waste

- Any excess movement that doesn't add value to the product, service or process
- Solution is to rearrange layouts to decrease the distance between stations and make it easier to reach things that are often used
- Causes include:
  - Poor process design and controls
  - Poor workstation/shop layout
  - Shared tools and machines
  - Workstation congestion
  - Lack of standards

Excess processing

- Often occurs due to the creation of multiple versions of the same task
- Solution is to standardize processes, empower employees and eliminate unnecessary documentation, sign-off processes and meetings
- Examples include:
  - Excessive reports
  - Multiple signatures
  - Re-entering data and duplicated data
  - Lack of standards
  - Poor communication
  - Overdesigned equipment
  - Misunderstanding customer's needs
  - Human error

Benefits of Waste Elimination

- Systematic elimination of DOWNTIME can result in:
  - Faster processes
  - Lower costs
  - Higher quality
  - Happier workers, and most importantly
  - Happier customers